

GSFANet: Global Spatial–Frequency Attention Network for Infrared Small Target Detection

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Abstract—Infrared small target detection (IRSTD) has progressed significantly in spatial-domain learning. However, single-frame images’ limited spatial semantics impair discrimination between targets and similar noise while complicating integrity detection of large-scale target. To address these, we propose global spatial–frequency attention network (GSFANet), which enhances the distribution difference between targets and noise from a frequency-domain perspective while preserving spatial information integrity. The core innovations consist of three modules: 1) parametric wavelet downsampling (PWD), preserving small target details during frequency refinement to prevent feature fragmentation; 2) hierarchical gated kernel attention (HGKA), capturing cross-level frequency relationships through cross-channel kernel attention (C^2K) and maintaining spatial coherence via cross-spatial gate attention (CSG), effectively bridging semantic gaps across layers; and 3) adaptive frequency-decoupled fusion (AdaFD), dynamically fusing target-associated frequency components while suppressing noise. We further develop AdaFL Loss to balance multiscale target gradients and stabilize training. Experiments on three benchmark datasets demonstrate GSFANet’s superior detection performance and enhanced segmentation robustness in complex scenarios compared with state-of-the-art (SOTA) methods. Our code will be made public at https://github.com/dengfa02/GSFANet_IRSTD

Index Terms—Cross-attention, infrared small target detection (IRSTD), spatial–frequency, transformer, wavelet transform (WT).

I. INTRODUCTION

INFRARED small target detection (IRSTD) delivers robust performance under complex illumination conditions, with applications spanning maritime surveillance [1], [2], remote sensing [3], precision guidance [4], and early warning systems [5], [6]. IRSTD approaches include multiframe methods

that leverage temporal–spatial correlations but suffer computational inefficiency [7] and single-frame detection offering superior efficiency with minimal prior knowledge requirements [8]. However, single-frame IRSTD faces critical challenges: 1) small and dim: targets appear small and dim due to long-range imaging, lacking textures and exhibiting low signal-to-noise ratios [9]; 2) multiscale: target sizes vary significantly (1–100+ pixels) based on distances and target properties [10]; and 3) shape ambiguity: blurred target boundaries complicate accurate segmentation, often incorporating background pixels that distort shape determination. Thus, achieving high-accuracy single-frame IRSTD remains challenging.

Earlier traditional approaches using image filtering [9], [11], human vision-based contrast enhancement [12], [13], and low-rank approximation [14], [15] relied on handcrafted features, limiting their adaptability to complex scenarios. Recently, some traditional methods based on improved tensor techniques [16] and sparse decomposition [17] have shown comparable performance to learning-based approaches in specific scenarios. Data-driven methods, particularly with multiscale encoding–decoding architectures, have emerged as promising alternatives. ACM [18] demonstrated benefits of asymmetric cross-layer feature fusion for preserving small target characteristics, inspiring enhanced fusion modules [19], [20], [21]. Other research bridged encoder–decoder semantic gaps [22], with DNA-Net [10] and SCTransNet [23] leveraging deep spatial semantics to suppress noise during decoding. Transformer-based architectures [3], [23], [24], [25] exploit global receptive fields for modeling long-range contextual relationships, reducing false alarms. Notably, MLP-Net [26] achieves similar global attention with lower computational cost. Additional gains have been realized through advanced loss functions [3], [27] and novel training paradigms [28].

A. Motivation

Current approaches primarily focus on spatial domain differentiation between small targets and noise. However, single-frame images contain limited semantic information, making this distinction challenging [19]. While deeper network architectures with multiscale encoder–decoder frameworks [3], [23] offer one solution, they introduce excessive branching that compromises inference speed.

1) *Why Frequency Useful*: In IRSTD, previous studies [29], [30] have observed that noise is typically distributed in the

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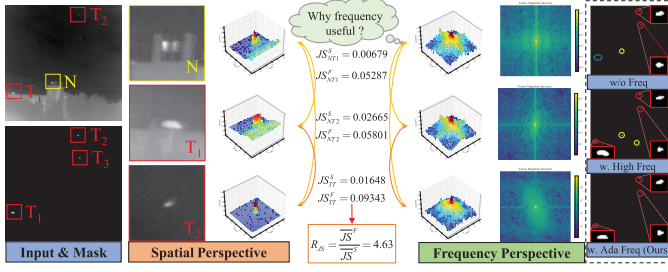


Fig. 1. Distribution differences of small targets in the spatial and frequency domains are compared. The last column, respectively, shows the detection results using the spatial domain, the inclusion of high-frequency components, and the proposed adaptive frequency approach.

high-frequency components of the image, while targets tend to reside in relatively low-frequency regions. Based on these findings, frequency information holds potential value for target modeling inIRSTD. However, there are still two notable limitations in prior studies [29], [30], [31]: 1) a lack of quantitative research regarding the frequency-domain distribution differences between targets and noise and 2) insufficient investigation of high-frequency noise that resembles target characteristics. Statistically, Fig. 1 illustrates a typicalIRSTD scene with high-intensity noise interference. Analysis of two targets T_1 , T_2 , and noise N reveals minimal spatial-domain differentiation, with Jensen–Shannon divergences of 0.00679 (T_1 – N), 0.02665 (T_2 – N), and 0.01648 (T_1 – T_2). However, frequency-domain analysis (by DFT) significantly amplifies these distribution differences (average gain $R_{JS} = JS^F / JS^S = 4.63$), highlighting the discriminative power of frequency-domain features. From a theoretical perspective, for a 2-D discrete image $I(x, y)$, the real part $\text{Re}(u, v)$ and imaginary part $\text{Im}(u, v)$ of 2D-DFT can be formulated as

$$\text{Re}(u, v) = \sum_{\cos \theta \geq 0} \cos \theta I(x, y) + \sum_{\cos \theta < 0} \cos \theta I(x, y) \quad (1)$$

$$\text{Im}(u, v) = \sum_{\sin \theta \geq 0} \sin \theta I(x, y) + \sum_{\sin \theta < 0} \sin \theta I(x, y). \quad (2)$$

Essentially, these components can be viewed as weighted contrast calculations among four nonnegative templates [32]. In essence, the wavelet transform (WT) can be considered as storing the spatial-domain contrast at multiple scales and orientations. With these contrasts,IRSTD in the frequency domain is statistically equivalent to searching for templates that best capture the differences between target and background. Such contrast information can be expressed more directly and efficiently in the frequency domain. From the perspective of sparse signal, target information is more concentrated and thus more easily detectable in the frequency domain. Furthermore, frequency-domain analysis naturally provides a global view, capturing both global statistical properties and local abrupt variations in the signal, which is theoretically advantageous for detecting sparsely distributed small targets.

The detection results in the final column in Fig. 1 illustrate these differences in practice. Methods relying solely on spatial modeling produces false alarms for noise N and misses target T_1 , and even high-frequency combined approaches with rigid frequency patterns still produce false positives.

In contrast, our approach incorporates WT for progressive frequency refinement with adaptive high/low-frequency decoupling. By leveraging learnable frequency-domain features, most background interference and noise can be effectively suppressed. We chose WT over DFT because it preserves crucial spatial–local structures during frequency decomposition. Unlike DFT, which eliminates local spatial relationships and remains dependent on input dimensions, WT enables precise multilevel operations and effective spatial–frequency information fusion capabilities essential for capturing small target features in complex scenes.

2) *Kernel Function*: Kernel functions in machine learning transform nonlinear problems into linear ones by computing dot products in high-dimensional feature spaces with reduced computational cost. Building on this idea, traditional cross-attention mechanisms essentially extract similarity coefficients from mapped features for weighted aggregation. Recognizing similarities between kernel functions and traditional cross-attention mechanisms, we substitute attention with kernel functions to avoid explicit feature mapping while enhancing high-dimensional nonlinear relationship capture (detailed in Section III-C).

3) *Feature Integrity*: Conventional downsampling methods present significant limitations: Maxpooling [10], [23] often confuses nonmaximum target edges with background, while strided convolution (SC) [21], though capable of learning some small-target features, struggles with extremely small targets, resulting in fragmented representations. Alternative approaches use lossless information transformation techniques through Haar WTs [33], [34] or pixel slicing methods in the spatial domain [19], [35], but single-domain downsampling inadequately addresses the dual challenge of enhancing crucial target structures while suppressing background noise. Our approach overcomes these limitations through complementary spatial–frequency-domain processing.

B. Contributions

To address the challenges mentioned above, we propose global spatial–frequency attention network (GSFANet), which decomposes spatial information into multilevel frequency components during encoding, progressively enhancing frequency resolution while reducing spatial resolution to construct a semantic pyramid. Global frequency attention and unified spatial attention mitigate encoder–decoder semantic gaps across all the levels, while an adaptive frequency decoder integrates adjacent levels for segmentation. Our main contributions are as follows.

- 1) We introduce a novel spatial–frequency joint learning framework that amplifies distributional differences between small targets and noise, enabling more accurate target recognition.
- 2) We propose a parametric wavelet downsampling (PWD) module that preserves critical spatial details of small targets through approximate spatial slicing, seamlessly integrating into the existing backbone architectures.
- 3) We design a global spatial–frequency attention mechanism based on kernel functions and gating mechanisms,

which significantly reduces computational complexity while ensuring unified target region focus across all the layers.

- 4) We propose a frequency-decoupled adaptive feature fusion module and adaptive focal loss that dynamically adjust cutoff frequencies and transition bandwidths, effectively balancing multiscale target features with gradient changes during training. This approach enables better adaptation to small target frequency characteristics across diverse scenarios while promoting stable convergence.

The remainder is organized as follows. Section II reviews relevant literature. Section III introduces GSFANet and its components. Section IV showcases the experimental results and analysis. Section V concludes with future directions.

II. RELATED WORKS

A. Attention-Based IRSTD Methods

IRSTD models have evolved beyond CNN-based networks [10], [36], [37] by incorporating attention mechanisms that establish global patch associations, broadly categorized as narrow and generalized approaches. Narrow attention mechanisms primarily use transformer-based architectures such as Swin Transformer [38], TCI-Former [28], and FTC-Net [39]. These approaches generally rely on CNNs to capture local features while incorporating transformers to model global contextual information. Advanced hierarchical structures in IRSTFormer [40], RKFormer [41], SCTransNet [23], and MTU-Net [3] leverage multiscale feature encoding and top-down feature aggregation to enhance detection performance through improved multilevel interactions. Generalized attention mechanisms use convolutional or gating techniques to emphasize spatial or channelwise information. For instance, ISNet [20] extracts precise target shape information through Taylor finite difference (TFD)-inspired edge blocks and two-orientation attention aggregation (TOAA), incorporating high-frequency joint learning. Similarly, IAANet [42] and MLP-Net [26] implement coarse-to-fine internal attention, modeling cross-layer associations between high-level semantics and low-level details. RDIAN [43] emphasizes spatial gradients in various directions using a multidirectional difference approach, which essentially acts as a form of spatial attention.

Unlike previous works, we improve conventional attention mechanisms through the use of kernel functions, establishing frequency associations between all the levels of encoders and decoders with low computational complexity. Meanwhile, we apply a gating mechanism to modulate global spatial representations, enabling efficient construction of global cross-channel and cross-layer semantic representations. This ensures the correct propagation of deep semantic features to low-level decoders, effectively reducing false alarms while maintaining low computational costs.

B. Deep Learning in Frequency Domain

While the existing approaches predominantly use spatial-domain features, frequency-domain analysis offers more

discriminative perspectives for IRSTD. WT facilitates multiscale frequency feature extraction by decomposing signals into different frequency components. Recent networks have integrated WT into neural architectures: Liu et al. [34] and Alaba and Ball [44] implement WT for downsampling in modified U-Net structures with inverse WT for upsampling, while Finder et al. [45] embed multilevel WT into convolutional layers to expand receptive fields without increasing parameters. In IRSTD, Zhu et al. [31] enhance discrimination between targets and background by applying fast Fourier transform to feature map patches, dividing and enhancing high/low-frequency features. However, their nonoverlapping patch approach compromises global frequency correlation modeling and target representation continuity. FreqODEs [46] use frequency-domain neural ordinary differential equations with frequency interaction attention to improve target-noise discrimination, introducing high-precision ODE solvers to suppress interference. However, this network lacks adaptive frequency selection capabilities for IRSTD. Both the approaches lack official implementation code and redefine test sets, preventing fair comparative evaluation.

Current frequency-based IRSTD methods remain underdeveloped, lacking sufficient hierarchical frequency feature interaction and adaptive frequency selection. Our method addresses these limitations by preserving spatial details during frequency-based downsampling and adaptively selecting relevant frequencies at high spatial resolution during decoding, significantly enhancing detection performance.

III. METHODOLOGY

This section presents the proposed GSFANet. Section III-A outlines the overall pipeline, followed by detailed descriptions and designs of specific components.

A. Overall Pipeline

Fig. 2 presents the GSFANet framework. The input image undergoes initial channel expansion via 1×1 convolution and ResBlock processing to produce the first-level encoder output $\mathbf{E}_0$ with C_0 channels. Addressing infrared small targets' multiscale characteristics, our PWD module performs $8 \times$ downsampling—an optimal balance considering that small targets typically span fewer than 50 pixels [3] and downsampling beyond $8 \times$ causes significant information loss [19]—while preserving target features and incorporating frequency information (channelized). The encoder architecture uses PWD modules and ResBlocks to progressively generate multiscale feature maps $\mathbf{E}_i \in \mathbb{R}^{C_i \times (H/2^i) \times (W/2^i)}$, ($i = 0, 1, 2, 3$). Overlap patch embedding with kernel size $P_i = P/2^i$, ($i = 0, 1, 2, 3$) and stride $P_i/2$ is applied to $\mathbf{E}_i$, ensuring continuous target modeling and producing embeddings $\mathbf{B}_i \in \mathbb{R}^{C_i \times (H_{eb}) \times (W_{eb})}$. Subsequently, the hierarchical gated kernel attention (HGKA) module thoroughly integrates frequency and spatial features, facilitating deep semantic propagation to lower layers for multilevel target focus. The resulting spatial-frequency mixed features $\mathbf{G}_i$ undergo size recovery through bilinear interpolation, convolution, and activation, followed by residual addition with encoded features $\mathbf{E}_i$

$$\mathbf{F}_i = \text{FR}_i(\text{HGKA}(\mathbf{G}_i)) + \mathbf{E}_i, (i = 0, 1, 2, 3). \quad (3)$$

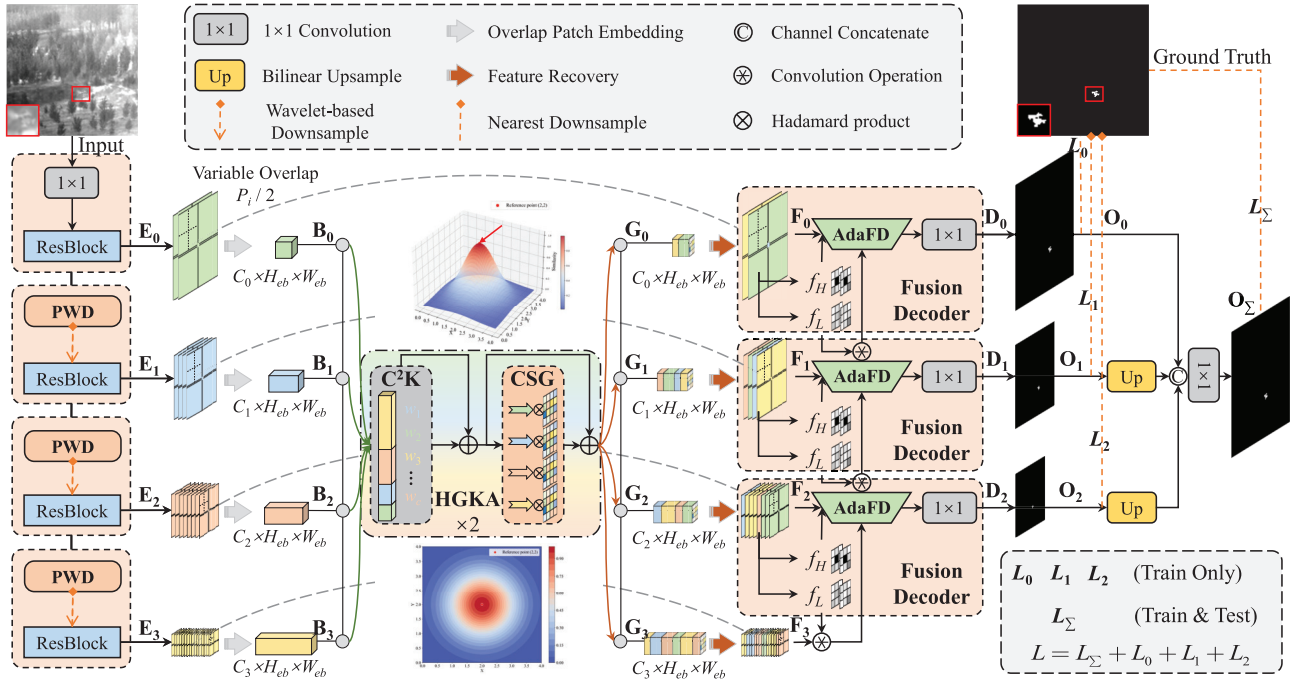


Fig. 2. Architecture of GSFANet achieves joint spatial–frequency learning through a frequency downsampling encoding process and an adaptive frequency fusion decoding process. It facilitates thorough mixing and interaction at each level of the encoder and decoder via the HGKA module.

For effective decoding of spatial–frequency mixed features $\mathbf{F}_i$, a fixed-frequency inverse WT would inevitably result in significant information loss. Therefore, the adaptive frequency-decoupled fusion (AdaFD) module generates n_k adaptive high-pass and low-pass filters simultaneously from low-level high-resolution feature maps. The high-pass filters sharpen low-level target details, while the low-pass filters extract precise target semantics from deep feature maps. The fused features pass through 1×1 convolution to generate the single-channel heatmap $\mathbf{D}_j$. Segmentation uses sigmoid activation with threshold $\tau_o = 0.5$ [23], yielding the result

$$\mathbf{O}_j = \mathbf{H}(\delta(\mathbf{D}_j) - \tau_o), (j = 0, 1, 2) \quad (4)$$

where $\mathbf{H}(x)$ denotes the Heaviside step function and δ denotes the sigmoid activation. During training, deep supervision (DS) [47] is implemented by upsampling $\mathbf{O}_1$ and $\mathbf{O}_2$ ($2\times$ and $4\times$, respectively), concatenating them, and applying convolution to produce $\mathbf{O}_\Sigma$. The loss is computed jointly for $\mathbf{O}_\Sigma$ and $\mathbf{O}_0$. The ground truth $\mathbf{Y}_0$ is downsampled via nearest neighbor interpolation to generate $\mathbf{Y}_j$ for supervision at different scales. During testing, only $\mathbf{O}_0$ is used as the final output.

B. Parametric Wavelet Downsampling

As discussed in Section I-A.1, both Fourier and WTs naturally serve as explicit regional contrast operators, projecting local image content onto a set of basis functions that emphasize spatial differences (i.e., templates). This property is advantageous forIRSTD, where targets are typically manifested as subtle, sparse regional contrasts within complex, noisy backgrounds. Conventional Haar WTs, while powerful, are limited in that their basis functions and decompositions are fixed thus lacking adaptivity to the diverse frequency

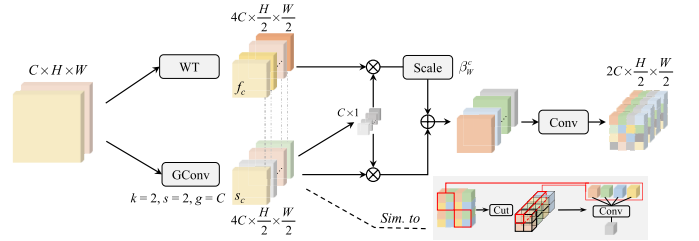


Fig. 3. PWD consists of a grouped convolution branch and a WT branch, which are aligned in both the frequency and spatial domains. It adjusts the decomposition frequencies of the target in a learnable manner while supplementing spatial details.

characteristics of real-world infrared targets and backgrounds. In the spatial domain, Cut-slice (or patch merge) operations are commonly adopted as downsampling modules in backbone networks. Yet, these methods lack explicit preservation of target-guided frequency and contrast information, potentially leading to loss of critical discriminative details. To address these limitations, the proposed PWD module introduces learnable frequency selection and spatial adaptivity into the wavelet framework:

In Fig. 3, the upper branch applies a fixed Haar WT, providing interpretable frequency decomposition as established in theory. The lower branch, inspired by the Cut-slice approach (mathematically equivalent to local spatial aggregation), leverages grouped convolution ($s = 2, k = 2$) to simulate WT-guided local contrasts and ensure spatial complementarity. The two branches are strictly channel-aligned for coherent fusion. Their outputs are fused using channelwise attention, generated via global context pooling and a shared fully connected (FC) layer. This adaptive weight-

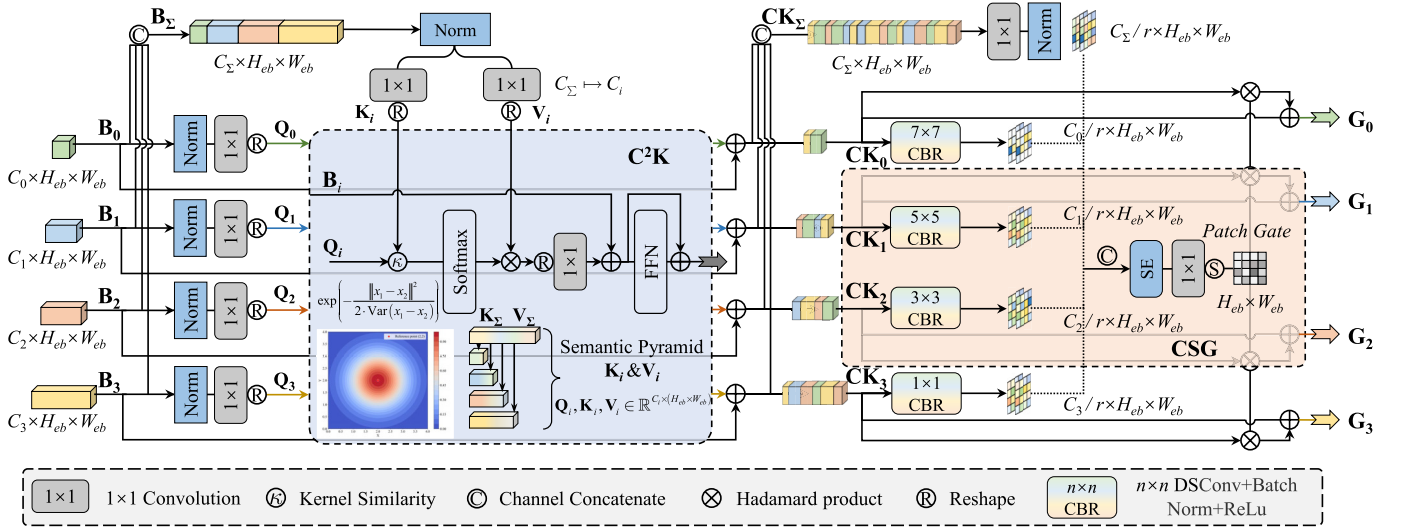


Fig. 4. HGKA module captures channel (i.e., frequency) relationships through cross-scale kernel attention and ensures alignment between deep semantic features and shallow localization features via globally consistent spatial attention.

ing allows the network to learn which frequency patterns or local contrasts best distinguish targets from background noise in each channel. Thus, PWD turns static templates into dynamic, data-driven frequency operators, leveraging the strengths of frequency domain analysis for IRSTD. As the network deepens, PWD enables progressive refinement of multiscale frequency features and supports global attention to maintain diverse, contrast-aware queries that enhance target-background separability.

We adopt the Haar basis for its intuitive local-contrast property and its balance of efficiency and detection performance. As shown in [45], Haar wavelets offer strong detection results with much lower computational cost than other bases.

C. Hierarchical Gated Kernel Attention

To effectively fuse frequency and semantic information across all the network levels, spatial transformers (STs) [3], [48] and channelwise transformers (CTs) [49], [50] serve as natural global mixers. However, using either approach in isolation results in insufficient interaction. While advanced architectures such as SCTransNet [23] and MetaFormer [51] incorporate joint spatial-channel interactions, they rely exclusively on spatial interactions between adjacent levels, potentially compromising small targets' global semantic consistency. Furthermore, traditional attention mechanisms' computational demands present significant challenges for IRSTD applications. To address these issues the following are put forth.

1) Based on the natural similarity between kernel functions and attention mechanisms, we integrate Gaussian kernel functions into the attention framework, enhancing nonlinear feature capture while reducing computation.

2) Recognizing that STs fundamentally establish inter-patch relationships, we transform these into patch-based gated attention mechanisms that maintain small target semantic consistency across all the levels while dramatically reducing computational costs.

The resulting HGKA effectively bridges the gap between STs and CTs with lower computational complexity, while ensuring comprehensive global fusion of frequency and spatial information throughout the network.

As shown in Fig. 4, given the input B_i at the i th layer, where $H_{eb} = \lfloor (H \cdot 2^{-i} - P_i)/(P_i/2) \rfloor + 1$, $W_{eb} = \lfloor (W \cdot 2^{-i} - P_i)/(P_i/2) \rfloor + 1$ ($\lfloor \cdot \rfloor$ representing floor function), comprises two components: cross-channel kernel attention (C^2K) and cross-spatial gate attention (CSG). The output maintains identical dimensions to the input, formally expressed as

$$B_\Sigma = \text{LN}(\text{Concat}[B_0, B_1, B_2, B_3]) \quad (5)$$

$$CK_i = C^2K(\text{LN}(B_i), B_\Sigma) + B_i, i = (0, 1, 2, 3) \quad (6)$$

$$CK_\Sigma = \text{BN}(f_{1 \times 1}(\text{Concat}[CK_0, CK_1, CK_2, CK_3])) \quad (7)$$

$$G_i = \text{CSG}(CK_0, CK_1, CK_2, CK_3, CK_\Sigma) + CK_i \quad (8)$$

where LN and BN represent layer and batch normalization, respectively, and $f_{1 \times 1}$ represents 1×1 convolution.

1) *Cross-Channel Kernel Attention*: C^2K establishes associations between single-level frequency inputs and all-level semantics. Traditional attention mechanisms present two critical limitations: 1) the need to explicitly project inputs into Q, K, V pairs, with each Q_i requiring dot product computation with K_Σ across all the levels, increasing computational burden, and 2) dot product operations (formally equivalent to linear kernels) inadequately capture high-dimensional nonlinear relationships in semantic space and fail to accommodate small target distribution characteristics.

To address these challenges, we introduce Gaussian kernel attention. Unlike conventional Gaussian kernels that require manual specification of variance σ and window size, our method adapts the kernel window to the input embedding shape at each scale (e.g., $C \times HW$), ensuring global context. More critically, σ is dynamically determined by the standard variance of the $Q - K$ vectors along the embedding dimension, allowing the Gaussian kernel width to adapt to the data distribution—a principle analogous to the $1/(d_k)^{1/2}$

scaling in classical attention. This unified design combines distribution adaptivity and similarity computation in a single kernel, removing fixed hyperparameters and aligning with the Gaussian assumption for small targets.

As shown in Fig. 4, input $\mathbf{B}_i$ directly generates matrices $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i$ through 1×1 convolution without additional 3×3 convolution, as Gaussian kernels inherently return the dot-product result in a possible infinite-dimensional feature space. We demonstrate this by expanding the kernel function e^{-x} using Taylor expansion

$$\begin{aligned} K(X_1, X_2) &= \exp\left(-\frac{\|X_1 - X_2\|^2}{\sigma^2}\right) = 1 - \frac{\|X_1 - X_2\|^2}{\sigma^2} \\ &+ \frac{\|X_1 - X_2\|^4}{2\sigma^4} - \frac{\|X_1 - X_2\|^6}{6\sigma^6} \\ &+ \dots = \sum_{k=0}^{+\infty} (-1)^k \frac{2^k}{k! \sigma^{2k}} \\ &+ \left(\sum_{k=1}^{+\infty} (-1)^{k+1} \frac{2^k}{(k-1)! \sigma^{2k}} \right) (X_1^T X_2) \\ &+ \left(\sum_{k=2}^{+\infty} (-1)^k \frac{2^{k-1}}{(k-2)! \sigma^{2k}} \right) (X_1^T X_2)^2 + \dots \quad (9) \end{aligned}$$

By definition $K(X_1, X_2) = \varphi(X_1)^T \varphi(X_2)$, for a four-dimensional sample $X = [a, b, c, d]^T$, the first few items are

$$\begin{aligned} \varphi(X) &= \left[\sqrt{\sum_{k=0}^{+\infty} (-1)^k \frac{2^k}{k! \sigma^{2k}}}, a \sqrt{\sum_{k=1}^{+\infty} (-1)^{k+1} \frac{2^k}{(k-1)! \sigma^{2k}}}, \dots \right]^T \quad (10) \end{aligned}$$

This high-dimensional mapping function contains infinite terms, significantly enhancing feature discriminability while circumventing distance measure failures in high-dimensional spaces through implicit mapping [52], [53]. The all-level feature $\mathbf{B}_\Sigma$ is to dimensionally match the current encoding level

$$\mathbf{Q}_i = \text{Flatten}_{H,W} (f_{1 \times 1}(\text{LN}(\mathbf{B}_i))) \quad (11)$$

$$\mathbf{K}_i, \mathbf{V}_i = \text{Flatten}_{H,W} (f_{1 \times 1}(\text{LN}(\mathbf{B}_\Sigma))) \quad (12)$$

where $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i \in \mathbb{R}^{C_i \times (H_{eb} \times W_{eb})}$, $\text{Flatten}_{H,W}$ denotes spatial dimension flattening (i.e., $\mathbb{R}^{C_i \times (H_{eb} \times W_{eb})} \mapsto \mathbb{R}^{C_i \times H_{eb} \times W_{eb}}$), and attention output is restored to original spatial dimensions postcomputation. The input then passes through a feedforward network (FFN) comprising two 1×1 convolutions. Our C^2K module is formally defined as

$$\mathbf{K}\mathbf{A}_i = \text{Softmax}\left(\exp\left(-\frac{\|\mathbf{Q}_i - \mathbf{K}_i\|_2^2}{2 \cdot \text{Var}(\mathbf{Q}_i - \mathbf{K}_i)}\right)\right) \odot \mathbf{V}_i \quad (13)$$

$$\text{FFN}(x) = f_{1 \times 1}(\text{ReLU}(\text{BN}(f_{1 \times 1}(x)))) + x \quad (14)$$

$$C^2K(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) = \text{FFN}(f_{1 \times 1}(\text{Re}_{H,W}(\mathbf{K}\mathbf{A}_i))) \quad (15)$$

where Var represents variance, $\odot$ denotes Hadamard product, and $\text{Re}_{H,W}$ restores spatial dimensions $\mathbb{R}^{C_i \times (H_{eb} \times W_{eb})} \mapsto \mathbb{R}^{C_i \times H_{eb} \times W_{eb}}$. Unlike traditional cross-attention mechanisms, C^2K selectively extracts the most relevant information from all-level features according to each layer's semantic level,

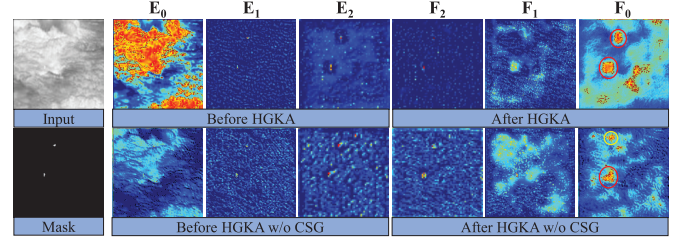


Fig. 5. AdaFD dynamically generates frequency-decoupled filters based on the frequency importance of feature groups, achieving finer frequency feature fusion through an efficient channel reordering strategy.

while its distance calculation naturally aligns with the Gaussian distribution characteristics of infrared small targets, significantly enhancing fine-grained semantic representation in multiscale feature hierarchical structure.

For computational analysis, vallina cross-attention in CTs [23], [49] applies linear projections to the i th layer input $\mathbf{B}_i \in \mathbb{R}^{C \cdot 2^i \times H_{eb} \times W_{eb}}$ and global feature vector $\mathbf{B}_\Sigma \in \mathbb{R}^{C_\Sigma \times H_{eb} \times W_{eb}}$ via channelwise (1×1) and spatial (3×3) convolutions, then computes weighted attention

$$\mathbf{Q}_i = \mathbf{B}_i W_q^{1 \times 1} W_q^{3 \times 3}, \mathbf{K} = \mathbf{B}_\Sigma W_k^{1 \times 1} W_k^{3 \times 3}, \mathbf{V} = \mathbf{B}_\Sigma W_v^{1 \times 1} W_v^{3 \times 3} \quad (16)$$

$$\text{CrossAtt}(\mathbf{Q}_i, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}_i \cdot \mathbf{K}^T}{\sqrt{d_k}}\right) \cdot \mathbf{V}. \quad (17)$$

In contrast, kernel attention requires only channelwise projection, maintaining $\mathbf{K}_i$ and $\mathbf{V}_i$ at the same dimensionality as $\mathbf{Q}_i$ [(11) and (12)]. The key difference lies in attention weight computation [(13) and (17)]. Conventional attention processes $\mathbf{Q}_i \in \mathbb{R}^{C \cdot 2^i \times (H_{eb} \times W_{eb})}$ and $\mathbf{K} \in \mathbb{R}^{C_\Sigma \times (HW)}$ with dot product Flops

$$\text{Flops}^{CA} = \sum_{i=0}^3 C \cdot 2^i \times H_{eb} W_{eb} \times C_\Sigma. \quad (18)$$

Kernel attention converts matrix multiplication to element-wise operations on $\mathbf{Q}_i, \mathbf{K}_i \in \mathbb{R}^{C \cdot 2^i \times (HW)}$ via two steps. First, $L2$ distances between row vectors require

$$\text{Flops}_1^{KA} = \sum_{i=0}^3 C \cdot 2^i \times H_{eb} W_{eb}. \quad (19)$$

Second, with attention dimensions reduce to $C \cdot 2^i \times 1$, computing squared terms yields total Flops

$$\text{Flops}^{KA} = \sum_{i=0}^3 C \cdot 2^i \times (H_{eb} W_{eb} + 1). \quad (20)$$

Therefore, kernel attention achieves approximately $1/C_\Sigma$ computational efficiency versus conventional attention while eliminating spatial projections and reducing Softmax complexity through smaller input dimensions.

2) *Cross-Spatial Gate Attention*: Previous approaches [19], [20] use progressive fusion to combine deep and low-level features. However, this method often causes deep-layer semantic dissipation during transmission, compromising small-target feature consistency. Fig. 5 (second row) visualizes attention heatmaps of HGKA inputs $\mathbf{E}_i$ and restored outputs $\mathbf{F}_i$. Without

globally consistent spatial attention, cross-level semantic information becomes inconsistent, preventing effective propagation of deep semantics $\mathbf{E}_2$ to low layer $\mathbf{F}_0$, leading to incorrect focus and degraded small target representation.

Leveraging the spatial consistency of embeddings $\mathbf{B}_i$, small-target features remain spatially aligned across levels. We therefore design a global gated mechanism to uniformly enhance target region responses across all the levels. As illustrated in Fig. 4, CSG processes the four-level outputs from $\mathbf{C}^2\mathbf{K}$, concatenating them along the channel dimension to obtain full-level feature $\mathbf{CK}_\Sigma$. As network depth increases, details progressively diminish while semantic information strengthens, making low-level features more susceptible to background noise. To address this, we apply convolution kernels with progressively decreasing sizes across different levels, effectively filtering low-level noise while reducing computational cost at deeper levels. The CSG process is expressed as

$$\mathbf{CK}_\Sigma = \text{BN}(\text{Concat}[\mathbf{CK}_0, \mathbf{CK}_1, \mathbf{CK}_2, \mathbf{CK}_3]) \quad (21)$$

$$\mathbf{g}_i = \text{ReLU}(\text{BN}(f_{(7-2i) \times (7-2i)}^{DSC}(\mathbf{CK}_i))), \quad i = (0, 1, 2, 3) \quad (22)$$

$$\mathbf{g}_\Sigma = \text{ReLU}(\text{BN}(f_{1 \times 1}(\mathbf{CK}_\Sigma))) \quad (23)$$

$$\mathbf{G}_\Sigma = \delta(f_{1 \times 1}(\text{SE}(\text{Concat}[\mathbf{g}_\Sigma, \mathbf{g}_i]))) \quad (24)$$

$$\mathbf{G}_i = \mathbf{CK}_i \odot \mathbf{G}_\Sigma + \mathbf{CK}_i \quad (25)$$

where $f_{(7-2i) \times (7-2i)}^{DSC}$ represents depthwise separable convolution with kernel size $7 - 2i$, and SE denotes squeeze-excitation channel attention. CSG confirms information importance using the original channel count, naturally emphasizing full-level and deep semantics without explicit level-specific weight assignment. As shown in Fig. 5 (first row), by generating globally unified gates, CSG ensures cross-level semantic consistency and effectively constrains attention to common focus areas, facilitating effective propagation of deep semantics $\mathbf{E}_2$ to low-level features $\mathbf{F}_0$. Furthermore, unified spatial attention shares gradients across layers during backpropagation, enhancing training stability.

D. Adaptive Frequency-Decoupled Fusion

After global mixing of small-target features via HGKA, conventional decoding using inverse WT with fixed frequency components [45] cannot fully exploit the complete frequency domain information. This limitation is particularly problematic for infrared small targets that exhibit critical characteristics across multiple frequency components. We address this by dynamically generating adaptive high-pass and low-pass filters based on frequency (i.e., channel) importance, enabling finer grained feature fusion.

As shown in Fig. 6, AdaFD conceptualizes the feature decoding process as a learnable filter generation task. Given globally mixed low-level feature $\mathbf{F}_i$ and high-level feature $\mathbf{F}_{i+1}$, depthwise convolution is performed on high-resolution features $\mathbf{F}_i$, dividing channels into n_k groups. The low-pass filter generator (LPF) evaluates each frequency component's contribution within a group (through average and Maxpooling), processes this through an FC layer, compresses the channels, and expands them to $9 \times$ the original size to generate

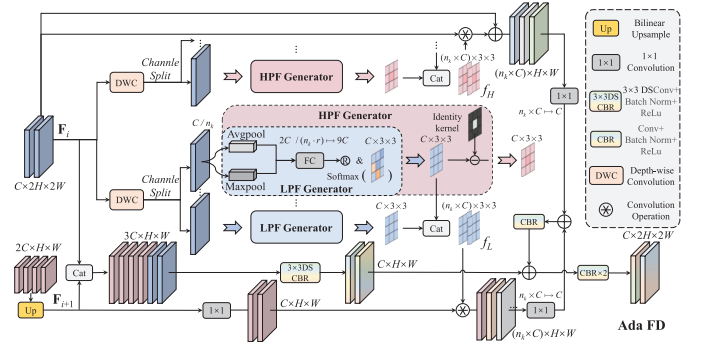


Fig. 6. Comparison results with and without globally consistent gated attention. It can be observed that globally consistent spatial attention facilitates the accurate propagation of deep semantic information to low-level features.

an initial filter. These extra channels are reshaped into spatial dimensions, yielding a $C \times 3 \times 3$ filter. A kernelwise Softmax operation transforms this into a low-pass filter [54]. This process is described as

$$\tilde{\mathbf{F}}_i^{n_k} = \text{Concat}[\text{Avgpool}(\mathbf{F}_i^{n_k}), \text{Maxpool}(\mathbf{F}_i^{n_k})] \quad (26)$$

$$f_L^{n_k} = \text{Softmax}(\text{Re}_{H,W}(\text{FC}(\tilde{\mathbf{F}}_i^{n_k}))) \quad (27)$$

$$\text{FC}(x) = f_{1 \times 1}(\text{ReLU}((f_{1 \times 1}(x)))) \quad (28)$$

where $f_L^{n_k}$ represents the low-pass filter generated for a group, $\text{Re}_{H,W}$ restores spatial dimensions, and $\mathbf{F}_i^{n_k}$ denotes grouped input features. Low-pass filters from all the groups are concatenated to produce the final low-pass filter $f_L \in \mathbb{R}^{(n_k \times C) \times 3 \times 3}$, which performs spatial convolution with deep features. This process adaptively determines which frequency characteristics of deep small targets should be preserved based on high-resolution features. By generating frequency-adaptive low-pass filters for each channel, LPF smooths and enhances internal feature consistency across adjacent levels, benefiting small target representation.

Similarly, after dividing $\mathbf{F}_i$ into n_k nonoverlapped groups, the high-pass filter generator (HPF) leverages residual connections to enhance details lost during downsampling. Following [55], [56], after generating the low-pass filter via LPF, a high-pass filter f_H is obtained by subtracting the low-pass filter from the identity kernel. This high-pass filter is applied through spatial convolution with low-level features. HPF restores downsampling-lost details by enhancing fine-grained target boundaries and adaptively strengthening important high-frequency components relevant to small targets. The enhanced high-frequency features are then added to target-aligned low-frequency features. After activation adjustment through a CBR block, the result is fused with concatenated compressed features. Finally, two additional CBR layers produce the adaptively decoded frequency results.

Unlike traditional fixed filters, AdaFD dynamically adjusts cutoff frequencies and transition bandwidths based on input feature distribution, enabling more precise frequency selection. This allows the model to automatically adapt to small targets' frequency characteristics across various scenarios, effectively focusing simultaneously on both fine target details and global semantic information.

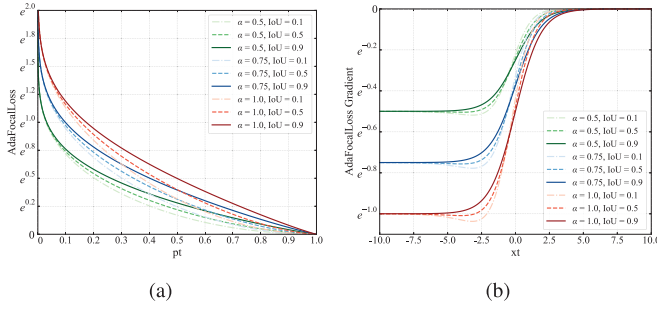


Fig. 7. Numerical analysis of AdaFL Loss (plot with $y = 1$). (a) AdaFL Loss curve. (b) AdaFL Loss gradient curve.

E. Adaptive Focal Loss

The IRSTD task confronts severe sample imbalance. While SoftIoU Loss [27] focuses on overall targets with reduced sensitivity to sample imbalance (making it popular in recent studies), it presents two significant limitations: 1) large-scale targets contribute more to IoU than small-scale targets, impeding gradient updates for small-scale targets, and 2) as training converges and IoU increases, the loss gradient paradoxically increases, causing late-stage performance oscillation and convergence issues. Formally, SoftIoU Loss is defined as

$$SL(p, y) = 1 - \frac{\sum p_i \cdot y_i + 1}{\sum p_i + \sum y_i - \sum p_i \cdot y_i + 1} = 1 - \frac{I}{U} \quad (29)$$

where p_i represents the predicted probability for the i th pixel, y_i is the ground-truth label, I is the intersection, and U is the union. The gradient is derived as

$$\begin{aligned} \frac{\partial SL}{\partial p_j} &= -\frac{(\partial I / \partial p_j) \cdot U - (I \cdot \partial U / \partial p_j)}{U^2} \\ &= -\frac{y_j \cdot U - (1 - y_j) \cdot I}{U^2} = \frac{I - y_j \cdot (U + I)}{U^2} \\ &= \begin{cases} -1/U, & y_j = 1 \\ \text{IoU}/U, & y_j = 0 \end{cases} \end{aligned} \quad (30)$$

Early in training, random predictions yield low IoU and high U , resulting in low gradients for both target ($y_j = 1$) and background ($y_j = 0$), slowing convergence. Conversely, at later stages of training, higher IoU and lower U cause gradients to increase, leading to performance oscillations.

While Focal Loss [57] addresses sample imbalance by emphasizing hard samples (small-scale targets, object edges, ambiguous targets), its static hyperparameters cannot adapt dynamically during training, and it lacks target integrity perception. To overcome these limitations, we propose AdaFL

$$\text{AdaFL}(p, y) = -\alpha \cdot y \cdot (1 - p)^{(1-\text{IoU})} \log p - (1 - \alpha) \cdot (1 - y) \cdot (1 - p)^{(1-\text{IoU})} \log(1 - p) \quad (31)$$

$$\alpha = \delta \left(1 - A_i^B / (\max \{A_i^B\} + 1) \right) \quad (32)$$

where α is the target size scaling factor, A_i^B is the i th target's area in a batch, and δ smoothens weights to handle large-scale differences in target sizes. As shown in Fig. 7(a), smaller targets have larger α , resulting in higher loss, thereby automatically balancing losses across different-sized targets.

In addition, as training progresses and IoU increases, the predicted p_i approaches stability, and loss diminishes smoothly. The AdaFL gradient is derived as

$$\begin{aligned} \frac{\partial \text{AdaFL}}{\partial p} &= \alpha y (1 - \text{IoU}) (1 - p)^{-\text{IoU}} \log p - \alpha y (1 - p)^{(1-\text{IoU})} \frac{1}{p} \\ &\quad - (1 - \alpha) (1 - y) p^{-\text{IoU}} \log(1 - p) \\ &\quad + (1 - \alpha) (1 - y) p^{(1-\text{IoU})} \frac{1}{1 - p} \end{aligned} \quad (33)$$

$$\frac{\partial \text{AdaFL}}{\partial x} = \frac{\partial \text{AdaFL}}{\partial p} \cdot \frac{\partial p}{\partial x} \quad (34)$$

where $p = \delta(x)$, and x represent the value predicted by GSFANet for each pixel. As shown in Fig. 7(b), small-scale targets receive enhanced gradient update opportunities, while the gradient decreases and stabilizes as IoU increases. For the 1-IoU exponent, when small-scale targets dominate the batch (i.e., lower overall batch IoU), the α larger and IoU smaller. As shown in Fig. 7(b), this further increases the gradients for these small-scale targets. Conversely, if large-scale targets dominate, the overall gradients decrease, but the α factor maintains sensitivity to small targets within the batch. In addition, gradient contribution primarily concentrates in $x_i \leq 0$ (i.e., $p \leq 0.5$), indicating that AdaFL maintains strong focus on hard samples. The network training loss function is

$$L = \sum_{j=0}^2 \text{AdaFL}(\mathbf{O}_j, \mathbf{Y}_j) + \text{AdaFL}(\mathbf{O}_\Sigma, \mathbf{Y}_0). \quad (35)$$

In summary, AdaFL adaptively adjusts gradients based on training phases, balancing multiscale target prediction while demonstrating enhanced contour awareness, particularly for blurred edges of small targets.

IV. EXPERIMENT

A. Evaluation Metrics

We evaluate the performance using the following metrics.

1) *Intersection Over Union*: IoU is a pixel-level metric that measures the model's ability to describe target shapes. It is defined as

$$\text{IoU} = \frac{A_{\text{inter}}}{A_{\text{union}}} = \frac{\sum_{i=1}^N \text{TP}[i]}{\sum_{i=1}^N (T[i] + P[i] - \text{TP}[i])} \quad (36)$$

where N is the number of samples, $\text{TP}[\cdot]$ represents the number of true-positive pixels, $T[\cdot]$ denotes the number of ground-truth positive pixels, and $P[\cdot]$ denotes the number of predicted positive pixels.

2) *F-Measure*: This is a pixel-level metric that balances precision (Prec) and recall (Rec) [21], providing a comprehensive evaluation. It is expressed as

$$F_1 = \frac{2 \times \text{Prec} \times \text{Rec}}{\text{Prec} + \text{Rec}}. \quad (37)$$

TABLE I

COMPARISON OF BASELINES ON THE SIRST, NUDT-SIRST, AND IRSTD-1k DATASETS. THE EVALUATION METRICS INCLUDE IOU (%), F_1 (%), P_d (%), AND F_a (10^{-6}). $\uparrow$ INDICATES HIGHER VALUES SIGNIFY BETTER PERFORMANCE, WHILE $\downarrow$ INDICATES LOWER VALUES SIGNIFY BETTER PERFORMANCE

Methods	SIRST(Tr=50%)				NUDT-SIRST(Tr=50%)				IRSTD-1k(Tr=80%)			
	IoU $\uparrow$	F_1 $\uparrow$	P_d $\uparrow$	F_a $\downarrow$	IoU $\uparrow$	F_1 $\uparrow$	P_d $\uparrow$	F_a $\downarrow$	IoU $\uparrow$	F_1 $\uparrow$	P_d $\uparrow$	F_a $\downarrow$
<i>Traditional methods (Unsupervised)</i>												
Max-Median [11]	4.172	10.67	69.20	55.33	4.197	7.635	58.41	36.89	6.998	8.152	65.21	59.73
Top-Hat [9]	7.143	14.63	79.84	1012	20.72	33.52	78.41	166.7	10.06	16.02	75.11	1432
IPI [14]	25.67	43.65	84.63	16.67	17.76	26.94	74.49	41.23	27.92	35.68	81.37	16.18
PSTNN [59]	30.30	39.16	72.80	48.99	14.85	35.63	66.13	44.17	24.57	37.18	71.99	35.26
MSLSTIPT [4]	10.30	18.83	82.13	1131	8.342	18.26	47.40	888.1	11.43	12.23	79.03	1524
TLLCM [12]	1.029	4.995	79.09	5899	2.176	7.225	62.01	1608	3.311	2.186	77.39	6738
WSLCM [60]	1.158	4.812	77.95	5446	2.283	5.987	56.82	1309	3.452	2.125	72.44	6619
HiLV-LRSD [17]	61.93	43.43	96.32	271.44	45.38	31.64	84.21	269.81	35.73	26.58	92.33	422.21
GCMUDO [61]	21.59	14.77	86.31	661.7	17.98	12.49	54.30	359.73	11.83	7.89	72.53	215.88
<i>Deep Learning methods (Supervised)</i>												
ACM [18]	68.355	81.203	90.458	29.851	68.971	81.636	96.402	12.754	64.217	78.235	90.572	9.205
AGPCNet [21]	67.724	80.757	93.893	26.219	81.302	89.687	96.931	12.156	65.151	78.923	89.899	26.684
DNA-Net [10]	77.317	87.208	95.038	15.649	<u>94.012</u>	<u>96.913</u>	98.624	3.102	66.335	79.787	90.909	16.359
ISNet [20]	69.537	82.032	95.419	31.879	71.379	83.314	95.238	34.332	63.481	77.685	91.582	22.888
MSHNet [47]	73.614	84.802	96.565	<u>14.829</u>	81.577	89.854	97.037	7.492	67.358	80.521	93.939	14.177
MTU-Net [3]	70.731	82.857	<u>96.947</u>	33.956	74.779	85.569	95.979	35.159	<u>67.421</u>	<u>80.563</u>	91.919	17.631
RDIAN [43]	68.080	81.009	94.656	84.726	83.497	91.006	98.201	14.156	64.022	78.090	89.226	13.684
SCTransNet [23]	<u>78.133</u>	<u>87.724</u>	95.419	16.547	94.005	96.909	<u>98.942</u>	5.125	64.687	78.584	91.919	<u>10.039</u>
UIU-Net [24]	68.245	81.126	93.511	19.785	85.617	92.251	98.201	<u>2.666</u>	65.690	79.318	91.246	13.475
GSFANet (ours)	79.597	88.639	97.710	11.360	95.949	97.933	99.259	1.356	70.521	82.737	<u>92.929</u>	20.174

3) *Probability of Detection*: P_d is an object-level metric that calculates the proportion of correctly predicted targets T_{pred} relative to all the targets T_{all}

$$P_d = \frac{T_{\text{pred}}}{T_{\text{all}}} \quad (38)$$

where T_{pred} accumulates the count of targets (not pixels) with centroid deviations less than 3 pixels [10].

4) *False Alarm Rate*: F_a is another object-level metric that computes the proportion of false-positive pixels T_{false} relative to all the image pixels P_{all}

$$F_a = \frac{T_{\text{false}}}{P_{\text{all}}} \quad (39)$$

where T_{false} sums all the pixels from incorrectly predicted targets exceeding the centroid threshold.

5) *Receiver Operating Characteristic*: The ROC curve [58] describes the tradeoff between true-positive rate (TPR) and false-positive rate (FPR) under different classification thresholds. Curves positioned closer to the top-left corner indicate superior performance. The area under curve (AUC) quantifies overall performance, with higher values signifying better.

In addition, model complexity is evaluated using parameter count (Params), floating-point operations (Flops), and frames per second (FPS).

B. Implementation Details

1) *Datasets*: We conducted comparative experiments on three well-known public datasets: SIRST [18], NUDT-SIRST [10], and IRSTD-1k [20], consisting of approximately

2800 images in total. To ensure fairness, we adhered to the dataset splits defined by [10] for SIRST and NUDT-SIRST (50% for training) and by [20] for IRSTD-1k (80% for training), respecting the split standards established by the dataset creators. In Table I, “Tr” denotes the ratio of the training set.

2) *Settings*: We used a U-Net architecture with four-level skip connections as the detection backbone [47]. Configuration parameters included: three downsampling levels, encoder channel numbers of 16, 32, 64, and 128, a basic patch size of 32, two HGKA blocks, and an FFN channel expansion factor of 2.66 [23]. To preserve the original image sizes, inputs for the SIRST and NUDT-SIRST datasets were cropped to 256×256 , and to 512×512 for IRSTD-1k. To mitigate overfitting and enhance robustness to camera tilt, random flipping and small-angle rotation ($\pm 3^\circ$) augmentations were applied during training. The Adam optimizer was used with an initial learning rate of 0.001, a batch size of 8, and 1500 training epochs. Learning rate was decayed to 10^{-5} using cosine annealing. Following [10] and [23], the prediction threshold for evaluation metrics was fixed at 0.5. GSFANet was trained in PyTorch on a single Nvidia RTX 4090 GPU, an Intel Xeon Platinum 8352S CPU, and a Linux server with 512-GB RAM. For comparison experiments, traditional methods are implemented following the settings in [16], [22], and [47], while learning-based methods are trained using their original parameter configurations. Most of the adopted methods are open source and can be found at <https://github.com/XinyiYing/BasicIRSTD>

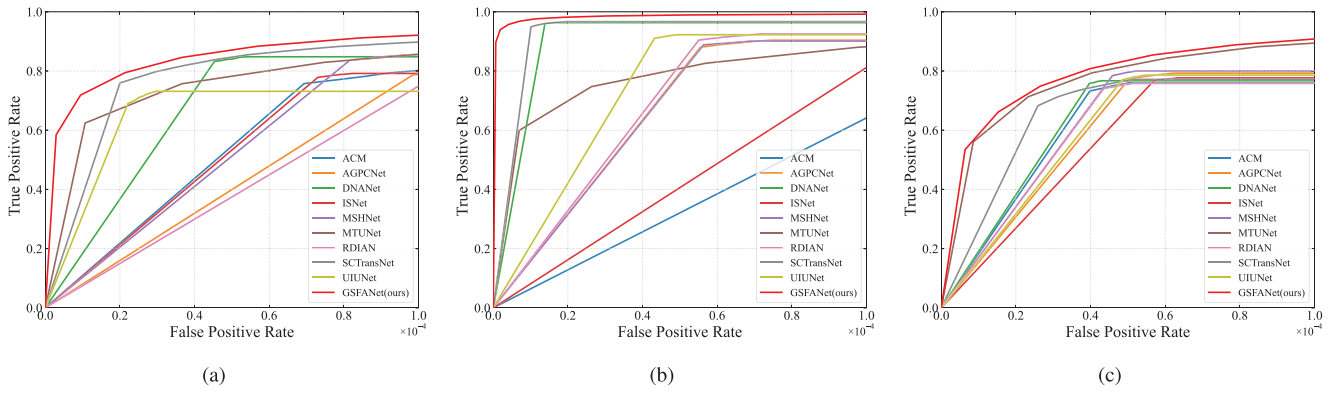


Fig. 8. ROC curves of different methods, where curves closer to the top-left indicate better performance. GSFANet achieves the best results on all three datasets: (a) SIRST, (b) NUDT-SIRST, and (c) IRSTD-1k.

TABLE II

COMPUTATIONAL COMPLEXITY AND PERFORMANCE OF COMPETITIVE METHODS. METRICS INCLUDE IoU (%), F_1 (%), P_d (%), F_a (10^{-6}), PARAMS (M), FLOPS (G), AND FPS (256×256). P_d AND F_a ARE EVALUATED ONLY FOR TARGETS $\leq 2 \times 2$ PIXELS

Model	Params ↓	Flops ↓	FPS ↑	IoU ↑	F_1 ↑	$P_d^{2 \times 2}$ ↑	$F_a^{2 \times 2}$ ↓
MTU-Net	8.221	6.197	114.3	70.977	82.996	78.125	5.087
DNA-Net	4.697	14.261	52.4	<u>79.221</u>	<u>87.969</u>	76.471	0
SCTransNet	11.191	20.240	37.8	78.942	87.739	82.353	0
UIU-Net	50.540	54.426	42.1	73.184	84.232	<u>85.259</u>	1.431
GSFANet (ours)	2.974	5.224	41.2	82.022	89.770	85.294	0.477

3) *Baselines*: We compared our method with nine state-of-the-art (SOTA) methods (MTU-Net [3], MSHNet [47], ISNet [20], DNANet [10], AGPCNet [21], RDIAN [43], SCTransNet [23], ACM [18], and UIU-Net [24]) as well as nine traditional methods (Max-Median [11], Top-Hat [9], IPI [14], PSTNN [59], MSLSTIPT [4], TLLCM [12], WSLCM [60], HiLV-LRSD [17], and GCMUDO [61]). Due to variations in dataset splits and training configurations among some models, we retrained these models under identical settings for fair comparison and report these results.

C. Quantitative Results

The quantitative results of GSFANet and baselines are shown in Table I. Overall, data-driven methods generally outperform traditional approaches due to limited generalization of handcrafted features across diverse scenarios. While HiLV-LRSD achieves comparable performance to data-driven methods in some cases, its dependence on local saliency and fixed low-rank decomposition rules causes imprecise target contour extraction, yielding lower F_1 scores. The proposed method achieves superior performance over SOTA methods on both the SIRST and NUDT-SIRST datasets, demonstrating its strong ability to distinguish between targets and noise as well as its robust shape extraction capabilities. Notably, while GSFANet does not achieve the lowest F_a on the IRSTD-1k dataset with its complex backgrounds, its enhanced frequency information enables more accurate contour detection of multiscale targets. Consequently, it significantly outperforms competing models in both IoU and F_1 metrics, highlighting its adaptability to diverse target characteristics across datasets.

TABLE III

COMPARISON OF AUC AT DIFFERENT THRESHOLDS. ↑ INDICATES HIGHER VALUES SIGNIFY BETTER PERFORMANCE

Model	SIRST		NUDT-SIRST		IRSTD-1K	
	$AUC_{\tau=0.5}$ ↑	$AUC_{\tau=1}$ ↑	$AUC_{\tau=0.5}$ ↑	$AUC_{\tau=1}$ ↑	$AUC_{\tau=0.5}$ ↑	$AUC_{\tau=1}$ ↑
ACM	76.923	80.344	75.219	81.254	72.831	74.418
AGPCNet	73.157	77.351	85.207	87.804	75.182	77.243
DNA-Net	80.883	82.850	95.003	95.674	73.615	75.162
ISNet	73.291	76.241	75.248	79.600	73.186	75.434
MSHNet	78.463	82.059	84.946	87.532	76.246	78.125
MTU-Net	<u>89.577</u>	<u>93.438</u>	90.591	93.541	<u>91.497</u>	<u>93.283</u>
RDIAN	74.692	79.485	87.284	89.906	72.415	74.116
SCTransNet	88.069	89.638	<u>95.668</u>	<u>96.172</u>	74.756	75.968
UIU-Net	71.403	72.278	88.212	90.236	74.719	76.673
GSFANet (ours)	93.368	95.008	99.007	99.117	91.642	93.695

We further compare GSFANet with high-performing MTU-Net, DNANet, SCTransNet, and UIU-Net. Table II presents the overall performance, evaluated as the average across three datasets. We further assess the models' performance on extremely small targets ($\leq 2 \times 2$ pixels). Since IoU metric is not appropriate for such small targets, we report P_d and F_a instead. Overall, all the models show a decrease in P_d when detecting extremely small targets. Models that aggressively suppress noise, such as DNANet and SCTransNet, achieve zero false alarms but inevitably suffer from low P_d due to missed detections. In contrast, GSFANet achieves a more balanced performance on extremely small target detection while also maintaining favorable model complexity. Notably, despite the GSFANet's smaller params and Flops, its inference FPS is not particularly high. This is primarily attributed to the custom wavelet decomposition lacking specialized CUDA optimization, resulting in suboptimal GPU bandwidth utilization (compared with MTU-Net's multibranch attention mechanism with dedicated parallel optimizations). Nevertheless, GSFANet still meets real-time detection requirements (≥ 30 FPS).

Fig. 8 illustrates the ROC curves for different methods across the SIRST, NUDT-SIRST, and IRSTD-1k datasets. GSFANet significantly outperforms other SOTA methods, achieving the highest TPR at low FPR, while demonstrating robustness to different segmentation thresholds. Furthermore, we quantitatively evaluated overall ROC performance using $AUC_{\tau=0.5}$ and $AUC_{\tau=1}$, corresponding to FPR thresholds of 0.5×10^{-3} and 1×10^{-3} , respectively, as shown in Table III.

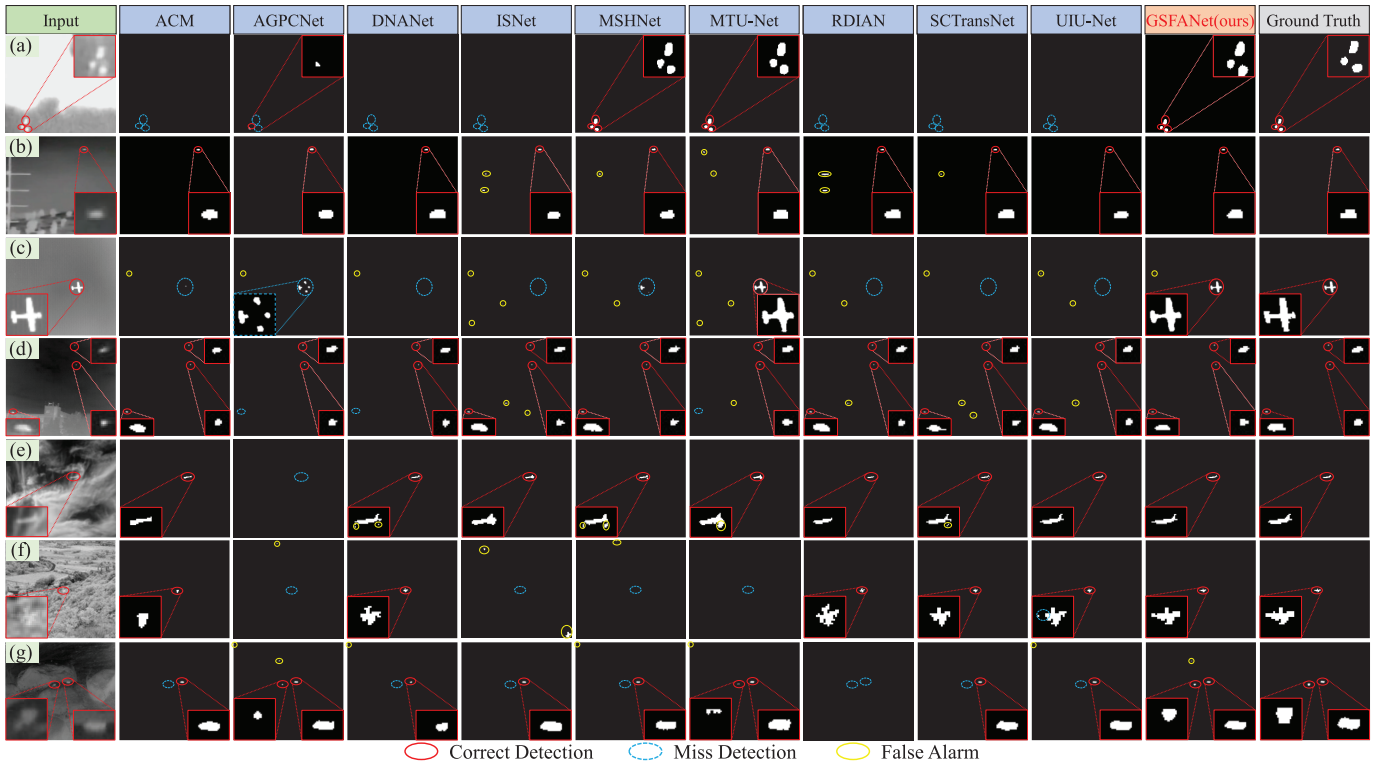


Fig. 9. Detection results of different methods, where red circles denote correct detections, blue dashed circles denote missed targets, and yellow circles denote false alarms. Closed-up views are shown in the corners. Images (a) and (b) are selected from SIRST. (c), (d), and (g) IRSTD-1k. (e) and (f) NUDT-SIRST. Examples of misdetections by the proposed method are also presented in (c) and (g).

Under both the thresholds, GSFANet achieves the highest AUC scores across all three datasets, demonstrating its superior adaptability.

D. Quantitative Results

To further demonstrate the superiority of GSFANet, Fig. 9 presents detection results of various SOTA methods on the datasets. The visual evidence clearly confirms our proposed method's exceptional performance. In example (a), most competing methods confuse targets with bright background regions, while only MSHNet, MTU-Net, and GSFANet successfully detect dense targets completely while avoiding interference from bright background areas. In cases with similar noise disturbances, such as (b) and (g), and especially (d), GSFANet successfully identifies the correct small targets, benefiting from the fine-grained frequency decomposition provided by PWD and the noise sensitivity of AdaFD. For extended targets, as seen in (c), (e), and (f), accurate boundary delineation and prediction completeness are especially critical. Leveraging the all-level HGKA frequency-spatial mixing and the boundary/semantic enhancement by AdaFD, our method better focuses on multiscale targets, propagates deep semantic cues, and boosts prediction confidence, thus accurately segmenting aircraft and drones. In summary, the proposed GSFANet delivers more precise small-target detection while outperforming SOTA models in both shape accuracy and completeness of extended target segmentation.

However, our method produces a false alarm in (g). As further analysis in Fig. 10 reveals, although some suspicious

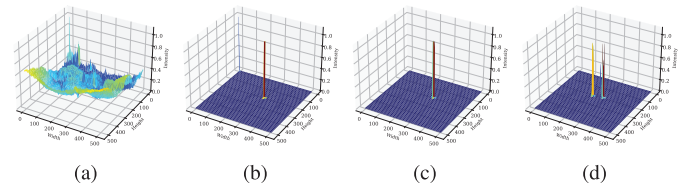


Fig. 10. Analysis of 3-D saliency maps (before thresholding) produced by different methods in Fig. 9(g): (a) input image, (b) DNA-Net, (c) SCTransNet, and (d) GSFANet. See zoomed-in view for a clearer view.

targets are partially suppressed during detection, AdaFD still generates high response values in these ambiguous regions. This limitation likely stems from insufficient granularity in frequency decomposition. In addition, all the methods generate false positives in (c), indicating the networks limited ability to capture semantics from a single frame. Incorporating multi-frame semantic association may further improve performance.

E. Ablation Study

1) *Integrated Framework*: We first evaluate the combined effect of all the modules by gradually removing the proposed components from GSFANet to verify their effectiveness, with experiments conducted on SIRST and NUDT-SIRST. Specifically, we sequentially exclude FEF, CSG, C2K, PWD, and DS; the results are shown in Table IV. Detection performance consistently improves as more proposed structures are incorporated. Notably, simultaneously removing C²K, CSG, and AdaFD causes the significant performance drop, demonstrating

TABLE IV

AVERAGE ABLATION RESULTS OF THE INTEGRATED FRAMEWORK ON SIRST AND NUDT-SIRST. THE EVALUATION METRICS INCLUDE IOU (%), F_1 (%), P_d (%), AND F_a (10^{-6})

GSFANet	+DS	+PWD	+C ² K	+CSG	+AdaFD	SIRST+NUDT-SIRST			
						IoU $\uparrow$	F_1 $\uparrow$	P_d $\uparrow$	F_a $\downarrow$
✓	✓	✓	✓	✓	✓	87.773	93.286	98.485	6.358
✓	✓	✓	✓	✓	✗	85.759	92.166	97.965	10.144
✓	✓	✓	✓	✗	✗	84.162	91.235	97.616	9.772
✓	✓	✗	✗	✗	✗	77.725	87.267	96.662	38.624
✓	✓	✗	✗	✗	✗	75.041	85.537	95.719	48.158
✓	✗	✗	✗	✗	✗	74.023	84.835	95.114	47.046

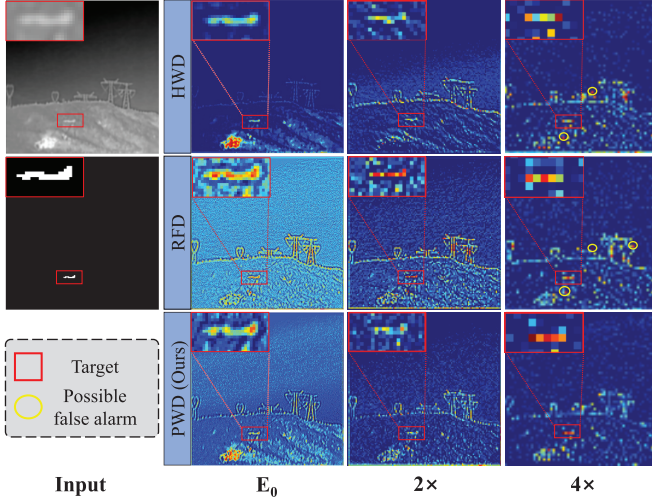


Fig. 11. Visualization of downsampling effects with HWD and RFD. The proposed PWD preserves more comprehensive target detail semantics while suppressing noise, maintaining better consistency within target regions.

TABLE V

ABLATION RESULTS OF DOWNSAMPLING MODULES ON SIRST AND NUDT-SIRST. THE EVALUATION METRICS INCLUDE IOU (%), F_1 (%), P_d (%), AND F_a (10^{-6})

Method	SIRST			NUDT-SIRST		
	IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$	IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$
Maxpool	79.357	96.183	14.509	95.712	99.153	2.620
SC	77.236	95.038	21.609	95.574	99.259	3.539
FPD [19]	79.584	96.565	22.659	95.812	99.153	1.930
RFD [35]	79.063	96.565	16.485	95.934	99.365	1.540
HWD [33]	79.447	95.038	22.165	95.921	99.365	3.056
PWD (ours)	79.597	97.710	11.360	95.949	99.259	1.356

the crucial role of HGKA's global spatial-frequency mixing in small target detection.

2) *Parametric Wavelet Downsampling*: To validate PWD effectiveness, we compared several common downsampling methods within GSFANet: Maxpooling, SC, RFD [35], HWD [33], and FPD [19]. Since PWD incorporates and unifies core concepts from HWD and RFD, no separate ablation on PWD alone was conducted. As shown in Table V, specialized downsampling modules generally outperform simpler alternatives; Maxpooling surpasses SC, enhancing target saliency, while SC compromises semantic integrity and dilutes small target signals in low-contrast scenes. Using only HWD or RFD is also inferior to PWD; Fig. 11 demonstrates that in 2 $\times$ downsampled feature maps, PWD preserves more accurate

TABLE VI

ABLATION RESULTS OF C²K ON SIRST AND NUDT-SIRST (INCLUDING MODEL COMPLEXITY). THE EVALUATION METRICS INCLUDE IOU (%), P_d (%), F_a (10^{-6}), PARAMS (M), AND FLOPS (G)

Method	SIRST+NUDT-SIRST			Params $\downarrow$	Flops $\downarrow$
	IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$		
vallina CA	87.226	97.721	14.522	5.441	5.806
linear kernel	87.385	98.241	11.781	2.974	5.251
Exp. kernel	87.568	98.294	14.876	2.974	5.251
Gus. kernel	87.773	98.485	6.358	2.974	5.251

aircraft detail than RFD. With increasing network depth, both RFD and HWD lose accurate airplane tail and introduce substantial noise that may cause false alarms. Through progressive frequency decomposition and adaptive spatial compensation, PWD preserves more integrated semantic information and critical details across all the levels.

3) *Cross-Channel Kernel Attention*: Gaussian kernel functions for implicit mapping and high-dimensional similarity extraction facilitate model lightweighting while enabling infinite-dimensional space dot products that amplify distinctions between small targets and noise. To demonstrate effectiveness, we compared standard cross-channel attention (CA), linear kernel, and exponential kernel functions. The results in Table VI reveal that kernel-based attention not only substantially reduces computation and parameters but also enhances performance compared with standard CA. Among these, Gaussian kernel cross-attention achieves optimal results due to two factors: first, compared with linear kernels, Gaussian kernels offer superior nonlinear expressivity and better modeling complex feature space distributions and second, the distance sensitivity and decay characteristics of Gaussians naturally align with infrared small target hotspot distributions and background statistical noise properties, enhancing target features in the attention mechanism while suppressing background interference, thereby improving discriminative capability.

4) *Cross-Spatial Gate Attention*: We use global cross-level gating to enhance semantic localization consistency for small target extraction. Our ablation study includes these variants.

1) *CSG w/o SE*: removes the SE mechanism, eliminating adaptive channel weight recalibration at each level before gate generation.

2) *CSG w UniConv*: applies uniform convolution kernel size across all the levels, preventing fine-tuned spatial noise processing for individual levels.

3) *CSG w SepGate*: generates separate gates for each level, preventing the network from focusing on common target locations across all the levels.

The results in Table VII demonstrate that since global C²K preceding CSG already provides sufficient channel-level semantic interaction, removing SE leads to only a slight performance drop. Conversely, using nonunified gates for each level results in the most significant performance decline. As illustrated in Fig. 5, nonunified gating fails to explicitly direct the network to focus on consistent locations and lacks integrated cross-level representation, leading to overall

TABLE VII

ABLATION RESULTS OF CSG ON SIRST AND NUDT-SIRST. THE EVALUATION METRICS INCLUDE IOU (%), P_d (%), AND F_a (10^{-6})

Method	SIRST			NUDT-SIRST		
	IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$	IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$
CSG w/o SE	79.427	96.565	15.065	95.920	99.259	1.563
CSG w UniConv	79.076	96.565	20.745	95.873	98.942	2.873
CSG w SepGate	78.324	96.183	21.177	95.589	99.259	2.022
CSG	79.597	97.710	11.360	95.949	99.259	1.356

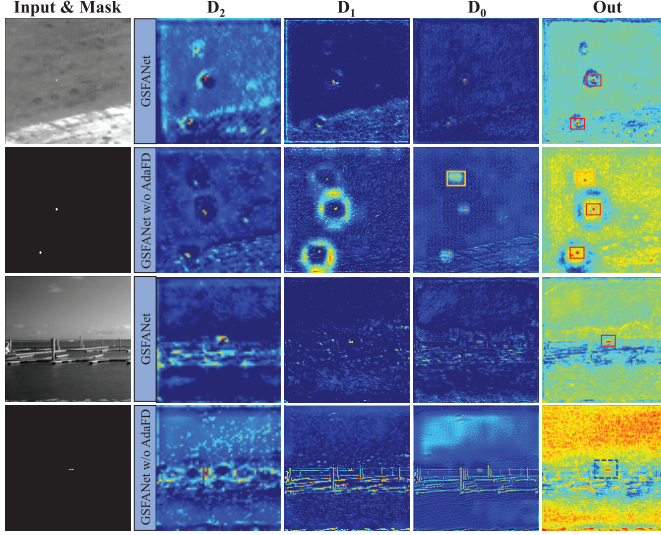


Fig. 12. Visualization of results with and without the AdaFD module. From left to right: input image, decoder layers 2, 1, 0, and final output. Red boxes denote correct detections, blue dashed boxes denote missed targets, and yellow boxes denote false alarms.

performance deterioration. Finally, applying uniform convolution kernel sizes across all the levels disregards the different semantic hierarchies, potentially introducing background noise in low-level features [19] and reducing global gate generation quality.

5) *Adaptive Frequency-Decoupled Fusion*: To recover information from mixed spatial-frequency representations, we decode final features through high-low frequency decoupling and fusion. To demonstrate effectiveness, we compared our method with three traditional fusion decoding approaches: CatFusion, FFD [19], and ACM Fusion [18]. In addition, we evaluated two variants.

1) *AdaFD w RI-HPF*: replaces AdaFD's low-level feature-map-based high-frequency convolution kernels with randomly initialized ones.

2) *AdaFD w/o f_H* : removes the high-frequency branch, generating only low-frequency kernels for spatial fusion.

As Table VIII shows, simple concatenation in CatFusion struggles to capture target-related frequency characteristics and integrate multiscale features effectively, resulting in lowest IoU. ACM improves overall performance by separately processing deep (channel-focused) and low (spatial-focused) features, but lacks adaptive channel frequency adjustment capability. FFD neither distinguishes between deep and shallow features nor provides flexible feature integration due to fixed fusion weights. In contrast, as illustrated in

TABLE VIII

COMPARISON OF ADAFD TO OTHER FUSION METHODS ON SIRST AND NUDT-SIRST. THE EVALUATION METRICS INCLUDE IOU (%), P_d (%), AND F_a (10^{-6})

Method	SIRST			NUDT-SIRST		
	IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$	IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$
CatFusion	77.612	96.947	10.434	94.826	99.153	2.987
ACM Fusion	78.548	96.565	22.350	95.557	99.259	1.999
FFD	78.652	96.947	22.165	94.845	99.471	1.517
AdaFD (Ours)	79.597	97.710	11.360	95.949	99.259	1.356

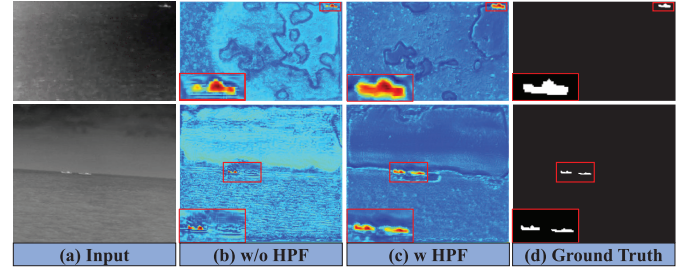


Fig. 13. Visualization comparison of decoder outputs with and without the HPF branch. The HPF branch ensures accurate and complete target boundaries in an adaptive high-frequency mode, while also preventing missed detection of adjacent targets. (a) Input. (b) w/o HPF. (c) w HPF. (d) Ground truth.

TABLE IX

ABLATION RESULTS OF ADAFD ON SIRST AND NUDT-SIRST. THE EVALUATION METRICS INCLUDE IOU (%), P_d (%), AND F_a (10^{-6})

AdaFD	$+f_L$	$+f_H$	$+RI$	SIRST+NUDT-SIRST		
				IoU $\uparrow$	P_d $\uparrow$	F_a $\downarrow$
RI-HPF	✓	✓	✓	87.608	98.103	9.784
AdaFD (Ours)	✓	✓	✗	87.773	98.485	6.358
AdaFD w/o f_H	✓	✗	✗	87.243	97.668	11.236

Fig. 12, AdaFD adaptively modulates deep- and low-level features using channelwise high- and low-frequency adjustment, enhancing high-frequency noise suppression and target semantic reinforcement, achieving superior performance among all the methods.

The AdaFD ablation results (Table IX) show that RI causes slight performance degradation, as it cannot leverage feature information to modulate necessary high-frequency components. When the high-frequency branch is removed, as depicted in Fig. 13, detection results with HPF (c) demonstrate more pronounced target ship contours compared with the results without (b) HPF, preserving structural consistency within contours and focusing heatmap responses more accurately on true target positions. In the second row, adaptive frequency-domain enhancement by HPF significantly improves distinction between targets and background wave patterns.

6) *Adaptive Focal Loss*: To validate AdaFL effectiveness, we trained GSFANet with Focal Loss [57], SoftIoU Loss [27], FocalIoU Loss [3], and AdaFL. The results in Table X confirm our analysis from Section III-D: SoftIoU Loss tends to yield poor performance due to undesirable gradient escalation in the later stages of training, hampering convergence. Although FocalIoU Loss combines elements from both the approaches,

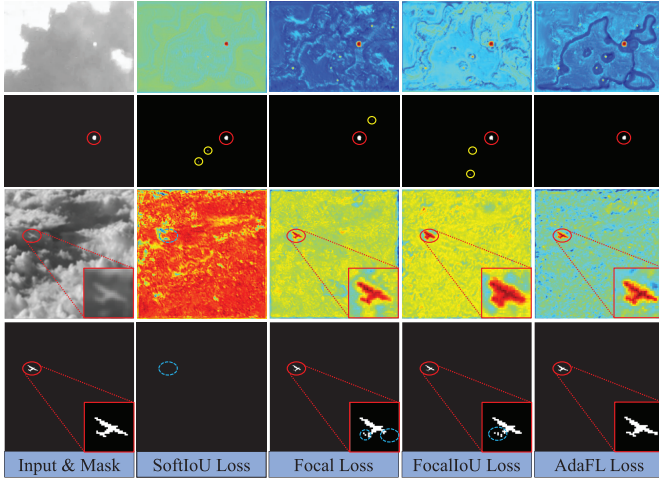


Fig. 14. Visualization results of different loss functions. The first row displays output heatmaps, while the second row shows threshold segmentation results. Red circles denote correct detections, blue dashed circles denote missed targets, and yellow circles denote false alarms.

TABLE X

COMPARISON OF ADAFL LOSS TO OTHER FUNCTIONS ON SIRST AND NUDT-SIRST. THE EVALUATION METRICS INCLUDE IoU (%), F_1 (%), P_d (%), AND F_a (10^{-6})

Function	SIRST				NUDT-SIRST			
	IoU $\uparrow$	F_1 $\uparrow$	P_d $\uparrow$	F_a $\downarrow$	IoU $\uparrow$	F_1 $\uparrow$	P_d $\uparrow$	F_a $\downarrow$
SoftIoU Loss	78.890	88.199	95.802	17.720	94.626	97.329	98.942	1.678
Focal Loss	78.051	87.672	96.183	14.818	95.176	97.529	99.259	4.251
FocalIoU Loss	77.544	87.352	95.802	29.204	95.562	97.731	99.365	1.609
AdaFL (Ours)	79.597	88.639	97.710	11.360	95.949	97.933	99.259	1.356

it still relies on Focal Loss, introducing hyperparameter sensitivity and lacking target scale awareness. Detection results with different loss functions are shown in Fig. 14. It can be observed that both Focal Loss and FocalIoU Loss focus excessively on suspicious targets at the pixel level, ignoring holistic target information leading to increased sensitivity to noise (first row mask) and insufficient perception of target integrity (second row mask). While SoftIoU Loss captures overall target structures, it struggles to distinguish between target boundaries and noise. AdaFL, by integrating IoU and adaptively adjusting gradients for targets and hard examples during training, both preserves target integrity and suppresses noise, achieving superior performance.

F. Hyperparameter Analysis

We investigated the effects of overlap degree, patch size, and the number of stacked HGKAs on performance. Specifically, L denotes the pixel overlap, P the patch size, and N the number of HGKA stacks. The results in Table XI show that performance improves as L increases from 0 to $P/2$, reaching optimal values at $L = P/2$. Larger overlaps reduce boundary-splitting risks for small targets, enabling more complete modeling. As P increases from 16 to 32, performance slightly improves; however, increasing to 64 yields no additional gains while raising computational complexity. We believe that excessively large patch sizes lead to overparameterization [45], resulting in overfitting to local contexts. Therefore, we set

TABLE XI

OVERALL INFLUENCE OF HYPERPARAMETER SETTINGS ON SIRST AND NUDT-SIRST. THE EVALUATION METRICS INCLUDE IoU (%), F_1 (%), P_d (%), F_a (10^{-6}), PARAMS (M), AND FLOPS (G)

Hyper-param	SIRST+NUDT-SIRST				Params ↓	Flops ↓
	IoU ↑	F_1 ↑	P_d ↑	F_a ↓		
<i>The degree of overlap</i>						
$L = 0$	85.724	92.061	97.340	20.012	2.974	4.953
$L = P/4$	87.172	92.903	97.202	9.309	2.974	5.019
$L = P/2^*$	87.773	93.286	98.485	6.358	2.974	5.251
<i>The size of patch</i>						
$P = 16$	87.668	93.228	98.103	8.115	2.188	5.859
$P = 32^*$	87.773	93.286	98.485	6.358	2.974	5.251
$P = 64$	87.609	93.185	98.538	9.751	6.120	5.079
<i>The number of HGKA</i>						
$N = 1$	87.062	92.848	98.156	7.509	2.570	5.160
$N = 2^*$	87.773	93.286	98.485	6.358	2.974	5.251
$N = 3$	87.490	92.488	97.965	16.190	3.378	5.342
$N = 4$	87.613	93.183	97.912	7.060	3.782	5.432

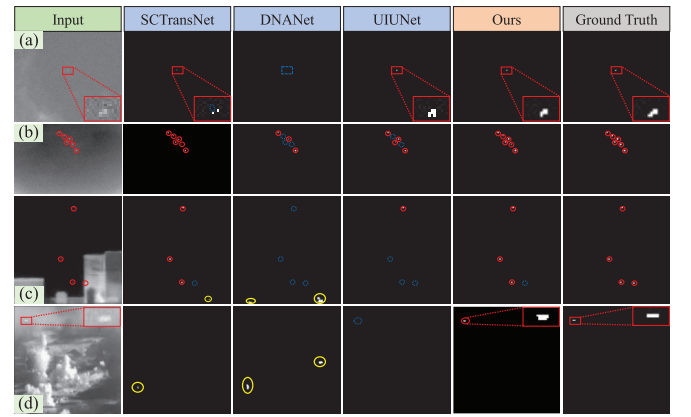


Fig. 15. Detection comparison under noisy conditions. Red, blue, and yellow boxes denote correct detection, miss detection, and false alarm, respectively. (a) and (b) Results under additive Gaussian noise. (c) and (d) Results under multiplicative speckle noise.

$P = 32$. For HGKA stacking, a significant performance drop at $N = 3$, attributed to excessive kernel attention stacking causing feature oversmoothing and reduced spatial precision, similar to attention collapse phenomena [62]. Performance partially recovers at $N = 4$, likely because the additional layer reconstructs suppressed details through gating mechanisms during channel concatenation, while overcoming depth-specific optimization challenges [63]. To balance efficiency and accuracy, we set $N = 2$.

G. Robustness

To further validate the GSFANet's effectiveness in detecting targets under complex noise conditions, we incorporated two common sensor noise types into the dataset: additive Gaussian noise and multiplicative speckle noise. Both the noise types were configured with zero mean and 0.001 variance [61] to simulate thermal and electronic noise typically encountered in infrared imaging sensors. These configurations emulate realistic image degradation scenarios under challenging imaging conditions.

As illustrated in Fig. 15, GSFANet maintains accurate target detection capability even in the presence of both the noise types. In Fig. 15(a) and (b), Gaussian noise introduces

significant interference within and around the target regions, causing DNANet and UIUNet, which lack global contextual awareness, as well as SCTransNet, which lacks comprehensive target integrity modeling, to produce miss detections. In Fig. 15(c) and (d), speckle noise impairs the models' understanding of building semantics. Although the proposed model exhibits one false negative in (c), its adaptive spatial-frequency hybrid design prevents false alarms, achieving superior performance compared with baseline methods. Overall, GSFANet demonstrates sustained target sensitivity under typical noise conditions, exhibiting robust performance in complex environments.

V. CONCLUSION

This article introduces the GSFANet to address two fundamental challenges in IRSTD: distinguishing targets from complex noise patterns and preventing fragmentation of large-scale target features. Our frequency-domain analysis reveals that this perspective significantly amplifies distributional differences between genuine targets and visually similar noise. The PWD achieves precise frequency decomposition while preserving detail features critical for detection. The lightweight HGKA module captures intrinsic frequency correlations and globally consistent semantic relationships, enabling effective cross-level feature interaction without the computational burden of traditional transformer architectures. Finally, the AdaFD accurately integrates target-related frequency components through dynamic filter generation, effectively suppressing background interference. Coupled with the gradient adaptive balancing mechanism of AdaFL Loss, GSFANet learns multiscale target features more stably during training, greatly improving detection accuracy and adaptability in complex scenarios.

GSFANet still has limitations: recognition errors persist in some complex single-frame scenarios, and computational complexity requires further reduction for practical deployment. Future work will explore temporal-spatial-frequency fusion mechanisms and implement optimization techniques including pruning and distillation to enhance deployment efficiency.

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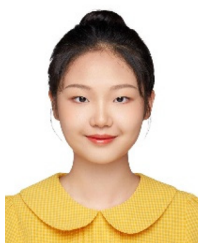


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